

Chapter 1

One shape

Draw a shape. Then change one thing about it, and see what happens.

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1.2 Add a square: predict, then reveal p.6
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One circle

This is a whole sketch. Run it, and a circle appears in the middle of the canvas.

```
1  import funground as f
2
3  f.size(600, 600)
4  f.background("oldlace")
5  f.no_stroke()
6  f.fill("tomato")
7  f.circle(300, 300, 200)
8  f.show()
```

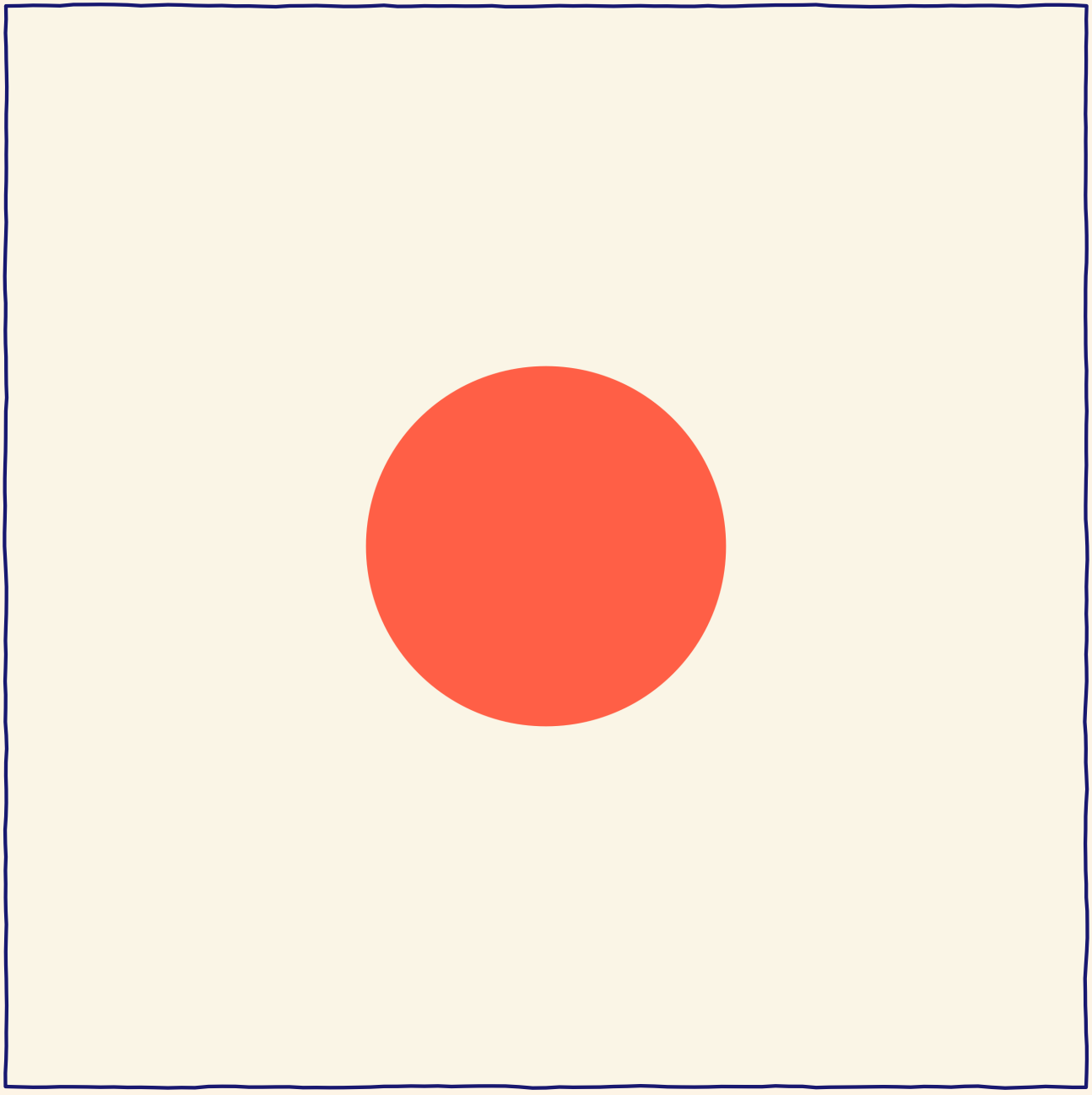
The canvas: 600 wide, 600 high.

A colour is a word, in quotes.

No outline: the shape is filled only.

300 across, 300 down: the middle.
200 wide.

Opens the window.

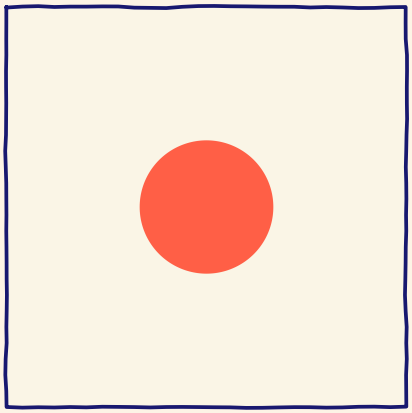


1.1 circle.py

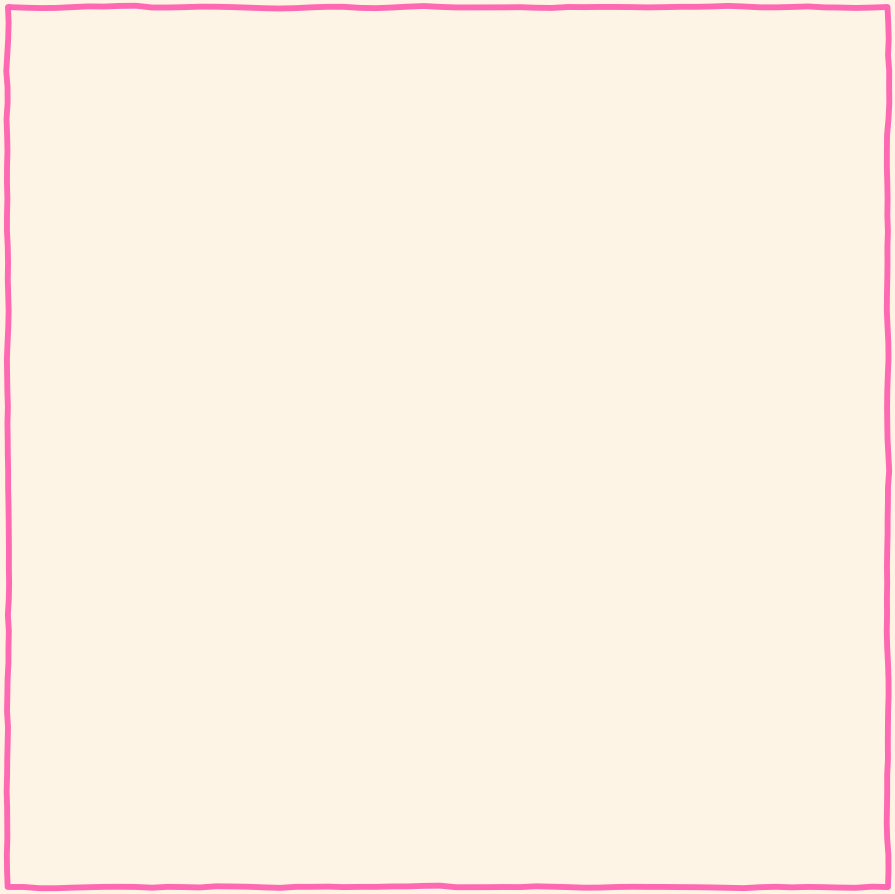
Add a square

Two new lines are in gold. Read them. What will the picture look like?

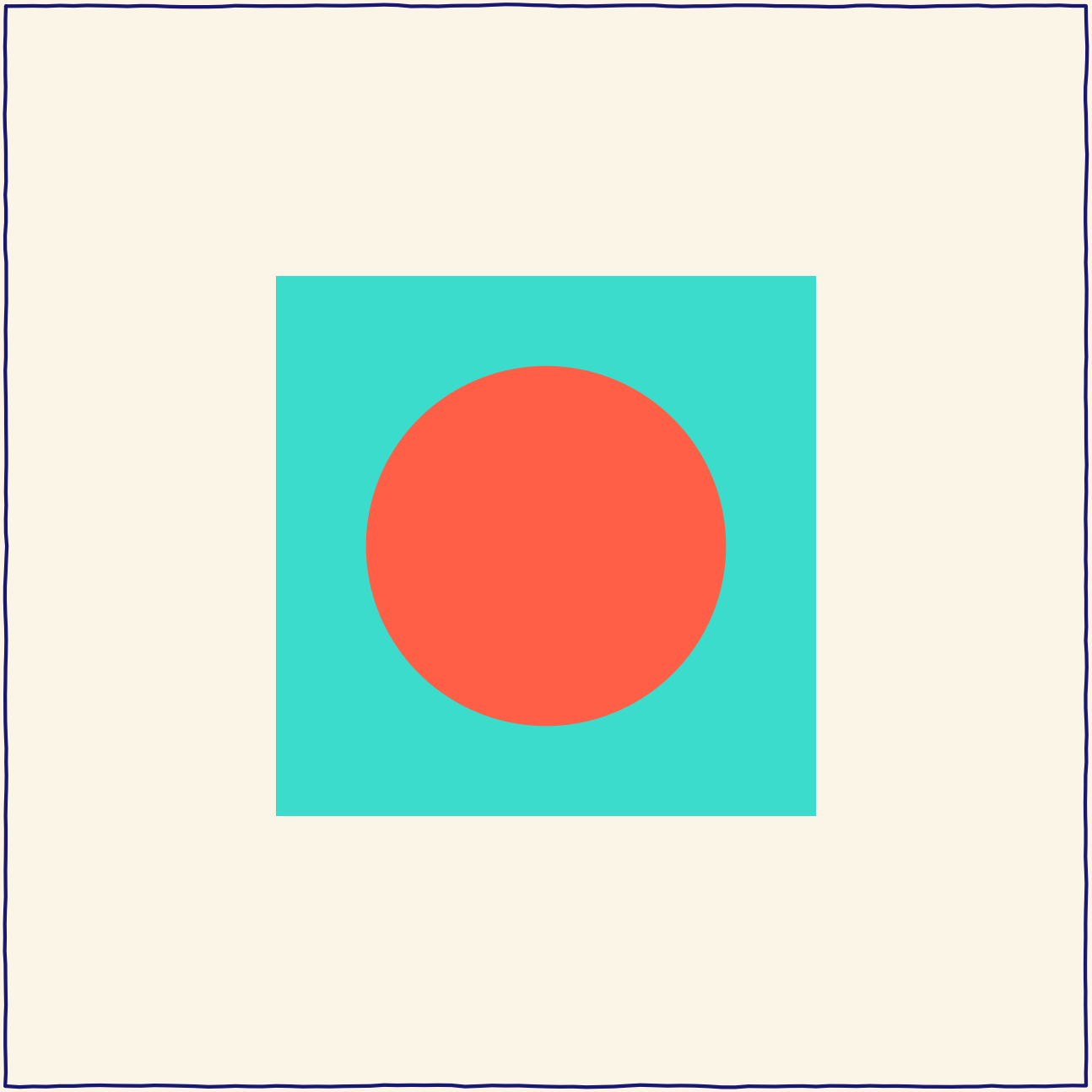
```
1  import funground as f
2
3  f.size(600, 600)
4  f.background("oldlace")
5  f.no_stroke()
6  f.fill("turquoise")
7  f.square(150, 150, 300)
8  f.fill("tomato")
9  f.circle(300, 300, 200)
10 f.show()
```



what you know: 1.1 circle.py



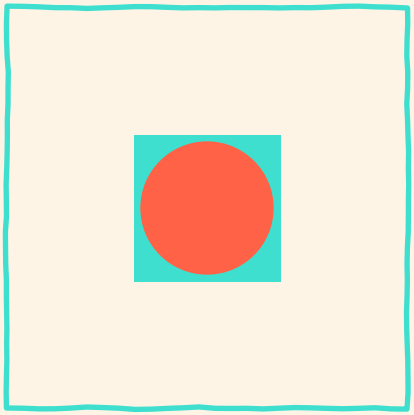
What will change? Draw it.



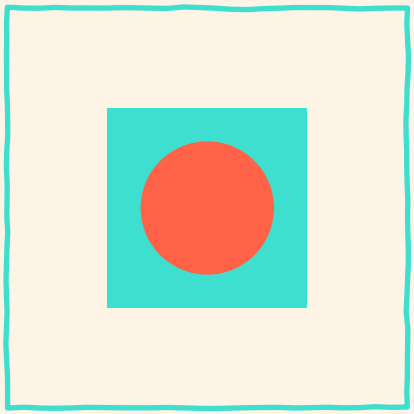
1.2 square.py

Change the size

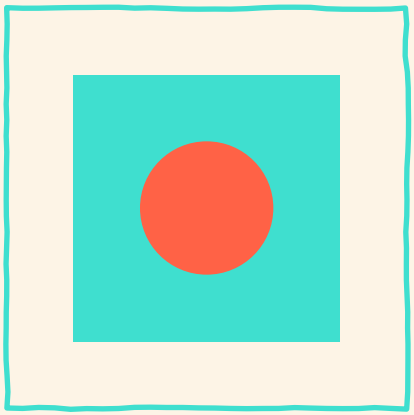
The square is 300 wide. Here are three sizes, side by side. Only the size changes.



size 220



size 300



size 400

Try this
Change 300 to 500 in the square. Where does it start now?

Somewhere random

f.random() picks a number, so the circle lands somewhere new. Each call picks its own place.

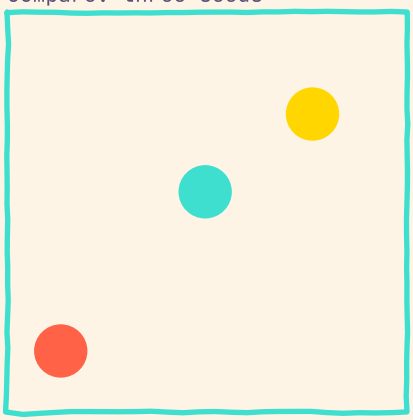
No outline: fills only.

Any place across, any place down. 80 wide.

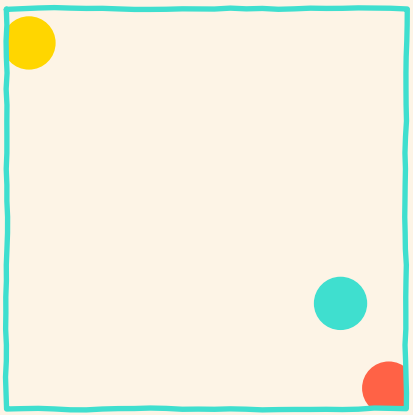
Same call. A new place.

```
1  import funground as f
2
3  f.size(600, 600)
4  f.background("oldlace")
5  f.no_stroke()
6  f.fill("tomato")
7  f.circle(f.random(600), f.random(600), 80)
8  f.fill("gold")
9  f.circle(f.random(600), f.random(600), 80)
10 f.fill("turquoise")
11 f.circle(f.random(600), f.random(600), 80)
12 f.show()
```

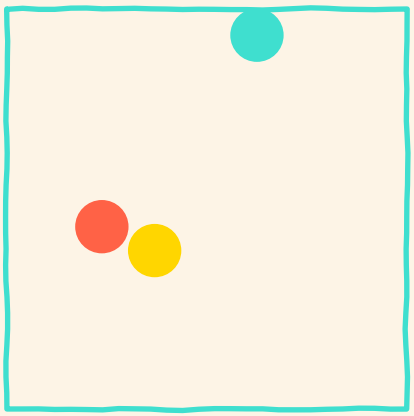
Compare: three seeds



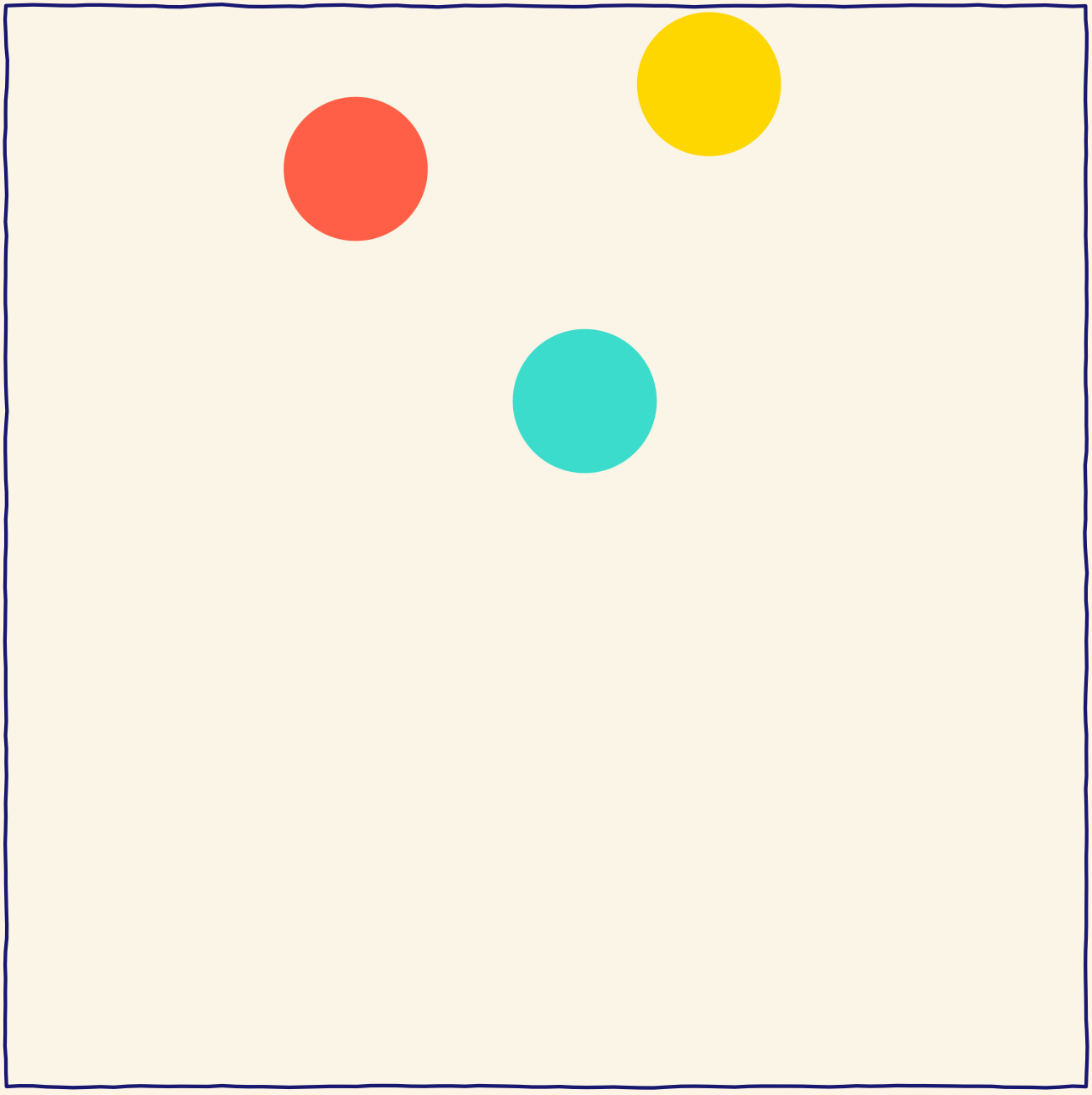
seed=1



seed=2



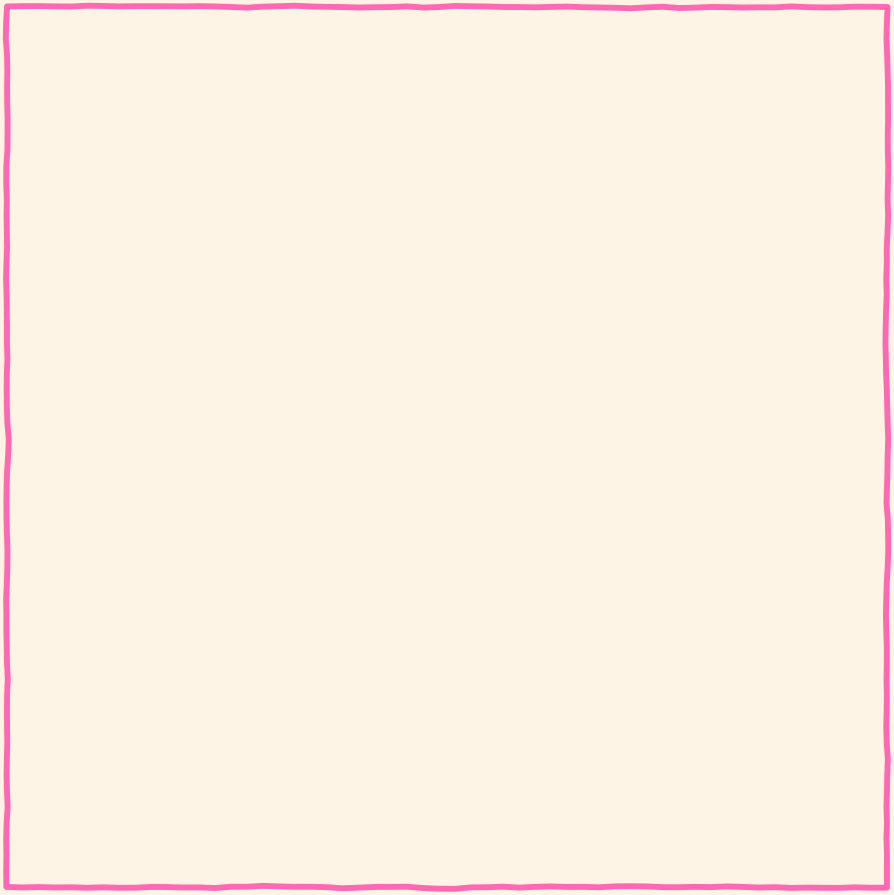
seed=3



1.3 random.py

Your page

Draw any shape. Change its colour, its size, and where it goes. Which change surprised you most?



Your canvas. Draw here, then try it in Play.

Colophon

Set in **Bricolage Grotesque**, Atkinson Hyperlegible Next and `JetBrains Mono`. All three are under the SIL Open Font License 1.1, which allows printing and embedding in a PDF.

The pictures are drawn by the sketches printed in this book. A program ran each one and placed its picture beside its code.

Made with funground. This chapter was laid out by a funground script: the grid, the type, the colours and the hand-drawn lines.

Paper is oldlace. Ink is midnightblue. The code in this book is CC0.